DSC 640 - Data Presentation and Visualization

Term Project - Milestone 2

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Purpose Statement:

The teachers' union aims to address the current teacher shortage by seeking state benefits for free childcare for staff; we recommend that teachers vote to authorize the union to propose funding for these childcare benefits to the state.

The Problem:

The high number of teacher vacancies in schools is causing current teachers to struggle with increased workload, less support, and diminishing resources.

The cost of childcare is rising steadily each year. Teachers fall into a category where they do not qualify for childcare assistance, but also do not make enough money where childcare is considered affordable.

The Solution:

To encourage more individuals to enter into the teaching profession and retain currently employed teachers, we recommend that the state provide full childcare benefits to school staff. We need teachers to vote "yes" to authorize the union to propose funding for these childcare benefits to the state.

Target Audience:

My Role:

I am a representative of the Florida Teachers Union. I am also a former veteran teacher with 13 years of experience, which gives me credibility in my position. The Florida Teachers Union is trying to get Florida teachers to support or vote/petition for the state to provide funding for free childcare for teachers as part of their state benefits. I have been put in charge of researching the rising childcare costs in the state. During my research I will also evaluate both the employment and teacher certification statistics in order to determine the approximate percentage of individuals who might be eligible to enter the teaching profession if they were provided free childcare. Then, I will be responsible for communicating and distributing what I have learned from the data to persuade teachers to vote to authorize the union to petition the state for full childcare benefits for teachers.

My Audience:

My audience is teachers who are both currently employed and members of the teachers' union. Teachers are well versed in data analysis as they use student and school data to drive their instruction. Data visualizations are frequently shown and analyzed during team meetings and meetings with administration. Using this data, I will need to show Florida teachers the benefits that free childcare could have on themselves, their colleagues and prospective teachers. Currently, many teachers are unhappy with the way the education system is being run and are struggling with increased workload due to staffing issues. Many teachers are fed up with the lack of support. They are seeking better resources, training, and mental health support to cope with the challenges posed by the shortage.

Visualization Mock-ups:

Mockups of the planned visualizations are included for each medium. Some of the data in the mockups is real because I have already started playing around with some of the visualizations in Power BI. But there is also data that is not real, and was created in MS Word "charts" to show the general type of visualization and where its placement will be on the medium.

Medium 1 - PowerPoint:

An in-person presentation given during the district union meetings. Attendees include union representatives as well as assigned teacher representatives from each school in the district. It will include all data findings and visualizations which support the recommendation that teachers vote to authorize the union to propose funding for these childcare benefits to the state. This PowerPoint will be available for each school's teacher representatives to take to their school should they wish to present the information at their next school meeting.

Design/Template Description: I chose to use the strategy of "lead with the ending" because teachers are constantly thinking about time management and are very critical of whether a not a meeting is worth their time. They will be interested in the "so what" right away, so I will use that to grab their attention. Then, I address the problem… in this case 2 problems: teacher shortages and rising childcare costs. The solution explains that by solving the problem of rising childcare costs, we can work to find a solution to the teacher shortage problem. A closing slide will review the benefits of teachers voting "Yes" to state funded childcare benefits for teachers and reiterates the call to action. As I am creating the PowerPoint, I may determine that the information planned for a single slide might need to be split into two slides in order to avoid too much clutter.

Medium 1 - PowerPoint Outline:

Slide 1 Title Slide: "Addressing Teacher Shortages Through State-Funded Childcare"

Slide 2: "Vote Yes: Empower Our Union to Secure State-Funded Childcare"

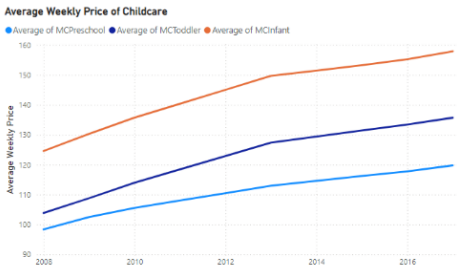
- Strategy: Lead with the Ending
  - Include the Purpose Statement with the Call to Action.

Slide 3: "Teacher Vacancies: How They Affect You"

- Introduce the Problem: Teacher vacancies
  - Points on how teacher vacancies are making current teacher's jobs more difficult.
- Multi-row Card: Displaying the number of teacher vacancies, and the total number of current teachers, the number of individuals that hold a teaching certification.
- Card: Showing the projected percentage increase in teacher vacancies for the 2024-25 school year.

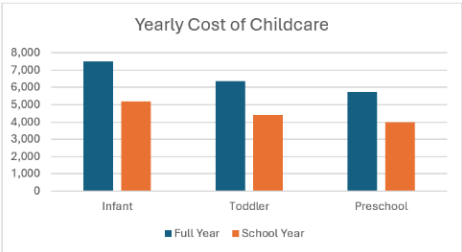
Slide 4: "The Steady Rise of Childcare Costs"

- Introduce the Problem: Rising childcare costs
- Line Graph: Displaying the rising childcare costs (for each age group) for the last 5 years.



Slide 5: "The Steady Rise of Childcare Costs" Continued…

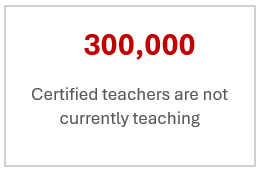
- Introduce the Problem: Rising childcare costs
- Teachers fall into a category where they do not qualify for childcare assistance, but also do not make enough money where childcare is considered affordable.
- Clustered Column Chart: Showing the average yearly cost of childcare for the 3 age categories.
  - "Full year" (52 weeks) is based on whether a teacher would need full year childcare benefits through the summer (because they teach summer school, have a



part-time job, or attend professional development). “School Year” (36 weeks) is based on teachers who only need childcare benefits during the school year.

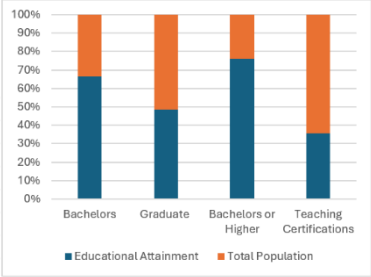
Slide 6: “Solving Childcare Funding: A Key to Easing Teacher Shortages”

- The Solution: Link the two problems together - How solving one problem can help alleviate the other
- Supporting state funding of full childcare benefits will strengthen teacher retention and encourage new teachers into the profession.
- Benefits of Teacher Retention:
  - Card: Displaying the current number of teachers with valid Florida teaching certificates, but not currently working in the field.



Slide 7: “Solving Childcare Funding: A Key to Easing Teacher Shortages” Continued…

- Benefits of encouraging more qualified individuals into the teaching profession:
  - 100% Stacked Column Chart: Depicting the percentage of people whose educational attainment includes the degrees eligible for applying for a Florida teacher certificate and how many individuals currently hold a teaching certificate, based on the 100% of the Total Population.



Slide 7: “Vote Yes: Empower Our Union to Secure State-Funded Childcare”

- Bulleted review the benefits that state funded childcare for teachers will have on the current teacher shortage.
  - Address w/ audience: I don’t have kids aged infant-preschool, so why should I care?
- Reiterate the Call to Action
- Encourage teachers to follow the FL Teachers Union Facebook group to keep updated with union activities.

Medium 2 - Handout:

A summarized version of the PowerPoint presentation that the teacher representatives from each school can distribute to teachers. In my teaching experience, the distribution of materials was very effective via teacher mailboxes.

Design/Template Description: The handout will have some, but not all of the visualizations from the PowerPoint presentation. It will use more concise language to highlight the main points of the presentation. I may decide to replace the “Yearly Cost of Childcare” column chart with a Multi-row Card (summarizing the same data) if space doesn’t allow for the full graph).

Medium 2 – Handout Template: (Single-sided handout)

<u>Medium 3 - Infographic/Website Post</u>:

The Florida teacher's union has a prominent social media presence for both Facebook and Instagram. An infographic posted on these social media groups would be an effective means of communication to the teachers on these platforms. The infographic will include only the key visualizations and data in order to simplify information, not overwhelm the audience.

Design/Template Description: I chose this infographic template because it allows the reader to view the visual with their natural left to right, z-pattern down to the bottom. The infographic will contain less visualizations such as tables and charts, and more cards and multi-row cards highlighting specific data only. This is because you want to catch the audience's attention as they are scrolling through the social media page. If they see a graph that will take them more than a few seconds to interpret, they will just keep scrolling.

Medium 3 - Infographic Template:

Vote Yes!

Addressing Teacher Shortages Through State-Funded Childcare

Empower Our Union to Secure State-Funded Childcare

The union needs your help...

Purpose Statement with the Call to Action

115% ⬆

Teacher vacancies are projected to rise from **4,776** in 2023-24 to **10,303** in 2024-25

High Teacher Vacancies

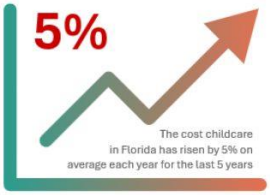
How They Affect You

Introduce the Problem: Brief bullet points on how teacher vacancies are making current teacher's jobs more difficult.

The Steady Rise of Childcare Costs

Teachers need childcare assistance

Introduce the Problem: Brief bullet points on how childcare costs are rising and how most teachers aren't eligible for assistance



300,000

Certified teachers are not currently teaching

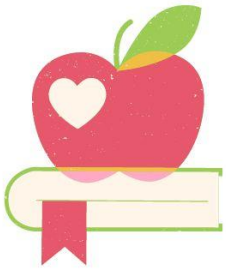
Solving Childcare Funding:

A Key to Easing Teacher Shortages

Link the two problems together - How solving one problem can help alleviate the other

Why should you care?

"Yes" to state funded childcare benefits everyone!



https://www.facebook.com/groups/floridateachersunion

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